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Lesson 3: Regression

Technologies for Artificial Intelligence

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Before we start

- Exercise 2, which we discuss now
- The pipeline you built is the one we fit models into from now on

Today: the first real model

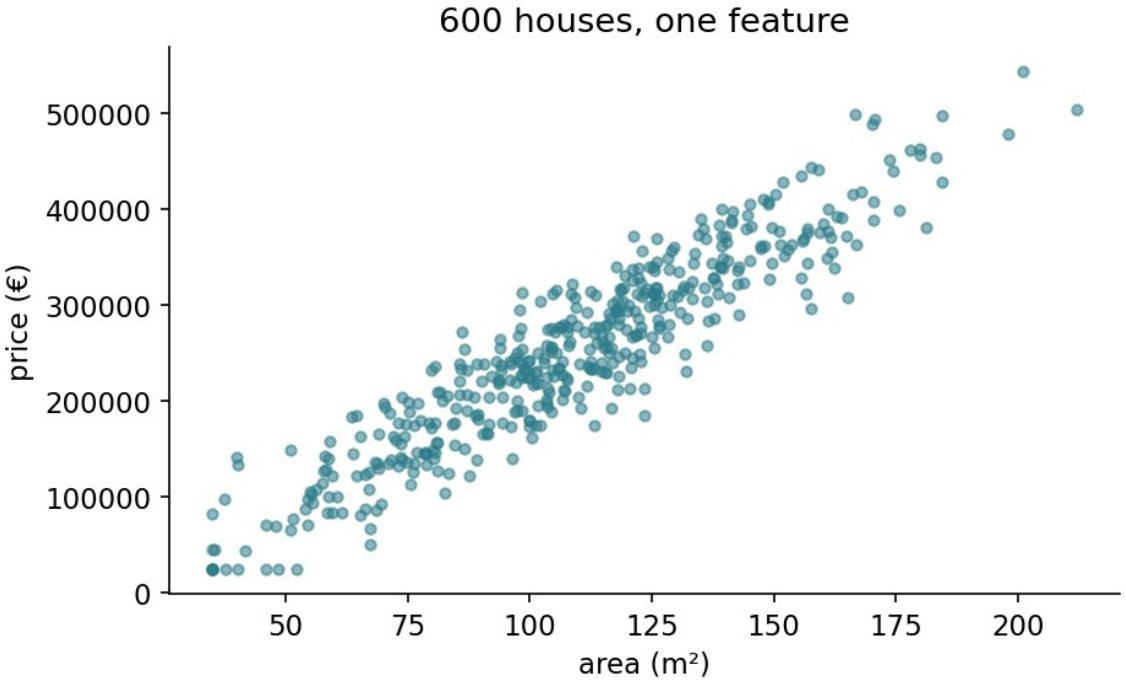
- The model and what it claims
- The exact solution, and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

One dataset, all lesson

600 houses. Six measurements. A price.

And we know the coefficients that generated it.

What we are trying to predict



What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up.

- Every square metre: the same 2,400 €
- Every kilometre out: the same −6,500 €

Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors fails: +50,000 and −50,000 cancel to zero.

Why squared, not absolute

- **Differentiable everywhere**: no corner at zero
- **Punishes one large error more** than several small ones
- Under Gaussian noise, it **is** maximum likelihood

The cost function: mean squared error

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

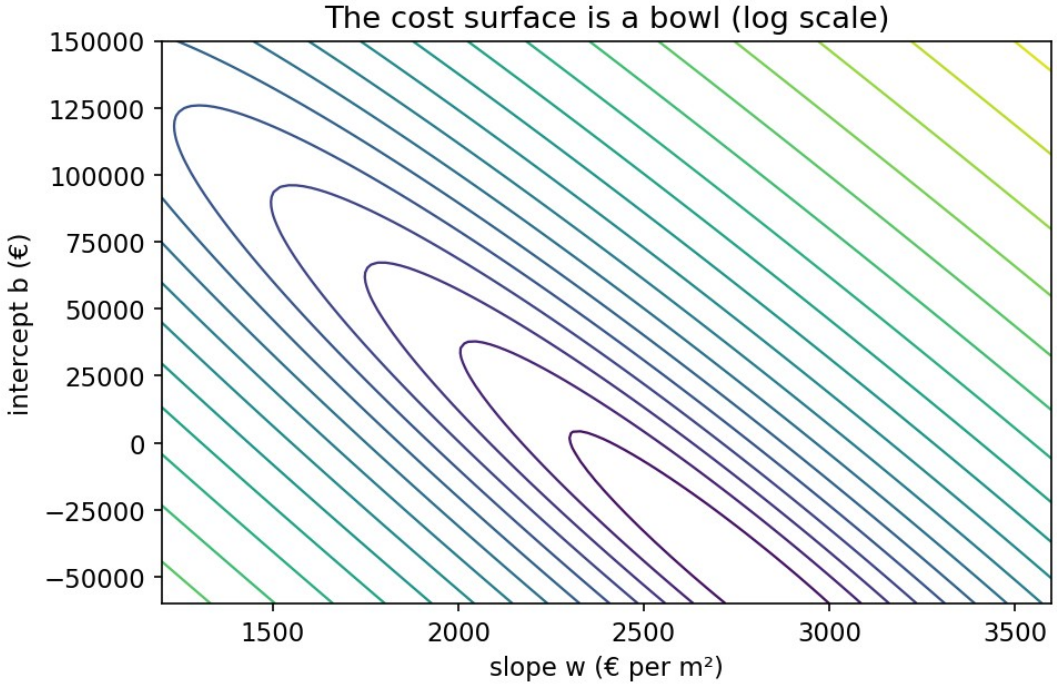
Trying $\hat{y} = 2{,}400 \cdot \text{area} + 45{,}000$

Area (m^2)	True (€)	Predicted (€)	Error $\hat{y} - y$	Error^2
80	240,000	237,000	−3,000	9.0×10^6
120	320,000	333,000	+13,000	1.69×10^8
200	540,000	525,000	−15,000	2.25×10^8

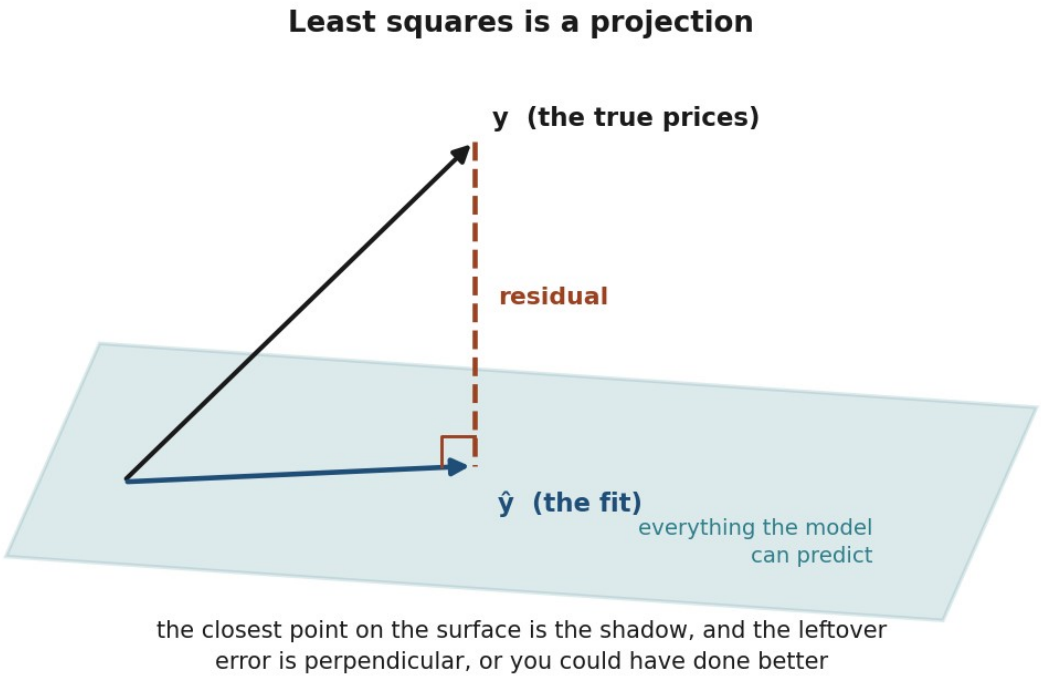
Squared error spends its attention on the worst house

- The 3,000 € error contributes about **2%** of the total
- The two large errors contribute the other **98%**
- Halving the 15,000 € error cuts the cost by **42%**

The cost is a bowl



The picture behind the exact solution



the closest point on the surface is the shadow, and the leftover error is perpendicular, or you could have done better

The normal equation

$$\theta = (X^\top X)^{-1} X^\top y$$

The same equation, by hand

Quantity	Value
the three houses	80, 120, 200 m² → 240k, 320k, 540k
X^TX	rows (3, 400) and (400, 60,800)
X^Ty	(1,100, 165,600)
determinant	22,400
slope w	**2,536 €/m²** (truth 2,400)
intercept b	28,600 €

And it really is the minimum

- The second derivative is X^TX, never negative in any direction
- So the cost is **convex**: one bottom, no local traps
- Any stationary point is *the* answer, not *an* answer

So why learn anything else?

- $(X^TX)^{-1}$ costs about n^3 operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

How stretched is the valley?

The **condition number** compares the steepest direction with the shallowest.

- Housing features as recorded: **285**
- The same features, standardised: **3.4**
- Add `area_sqft` beside `area_sqm`: **2,286**

The picture behind gradient descent

You are on a hillside in fog.

You cannot see the valley, but you can feel which way the ground slopes.

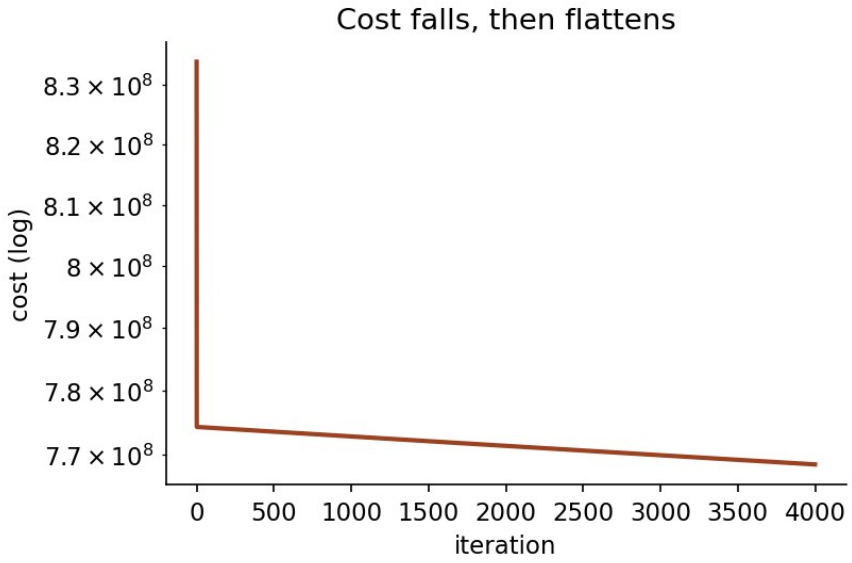
The update rule

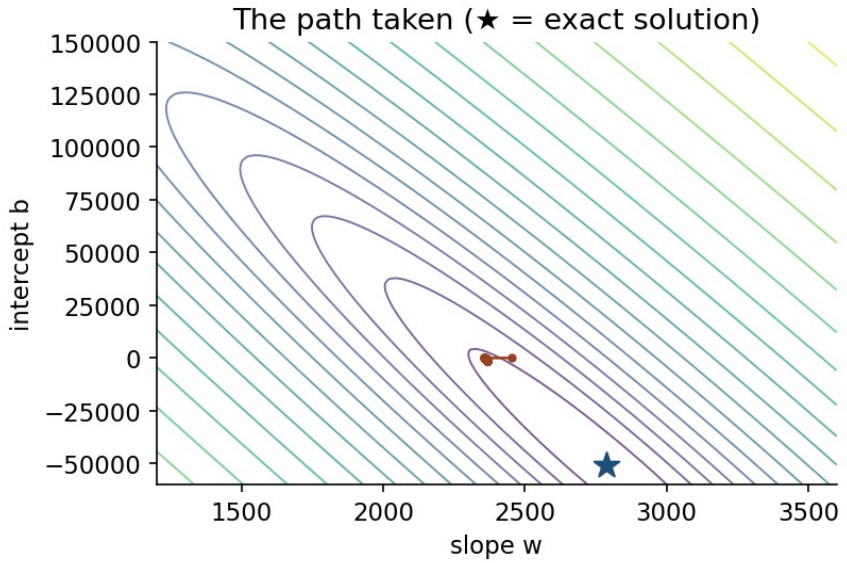
$$w \leftarrow w - \alpha \frac{\partial J}{\partial w}, \qquad \frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_i \left(\hat{y}^{(i)} - y^{(i)} \right) x_j^{(i)}$$

One step, from w = 0 and b = 0

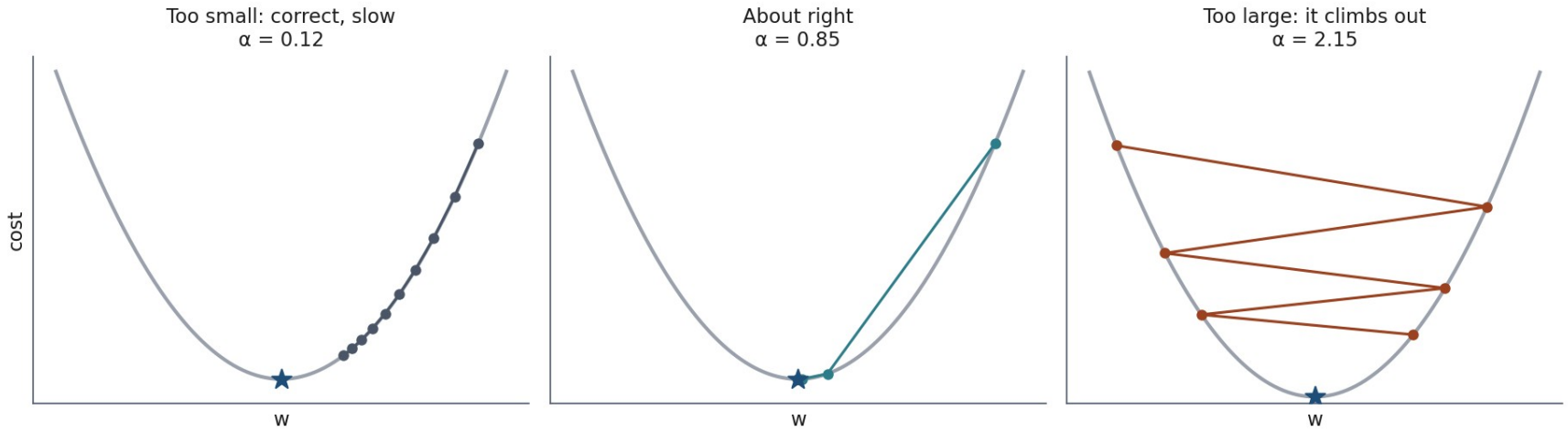
$\alpha = 10^{-5}$	Gradient	After one step
slope w	−63,600	0.636
intercept b	−390	0.0039

The path it takes





Three learning rates



Notebook 1, live

The model, the cost, the exact solution, the iterative one: from scratch.

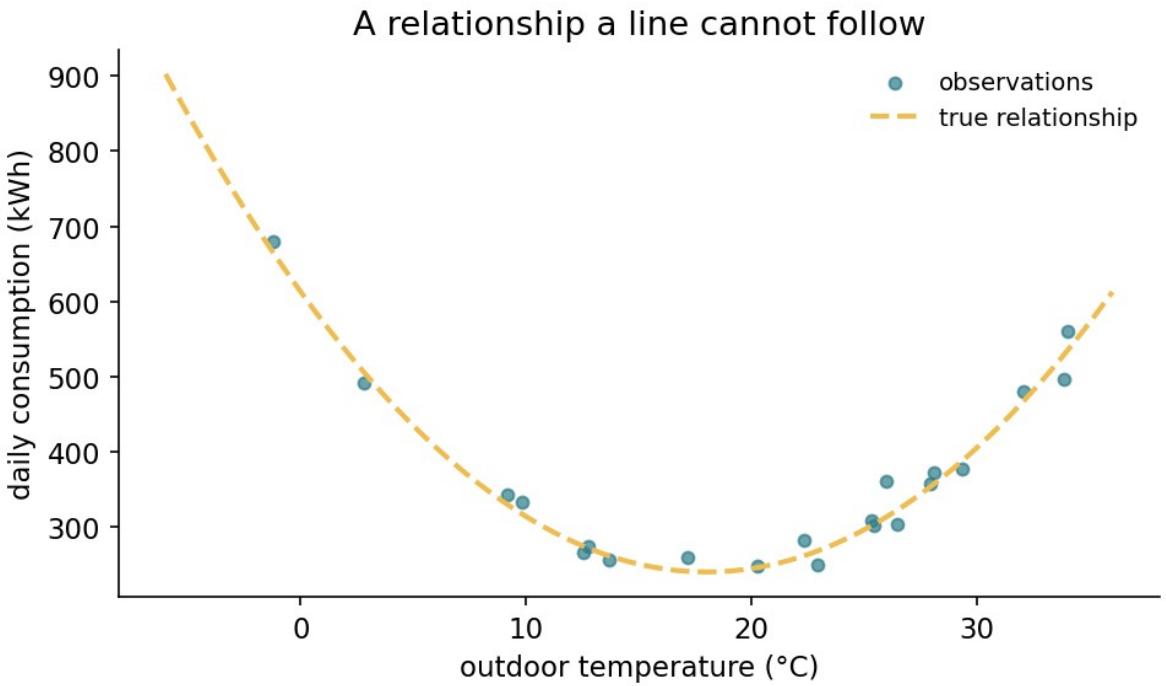
Break

What if the relationship bends?

Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

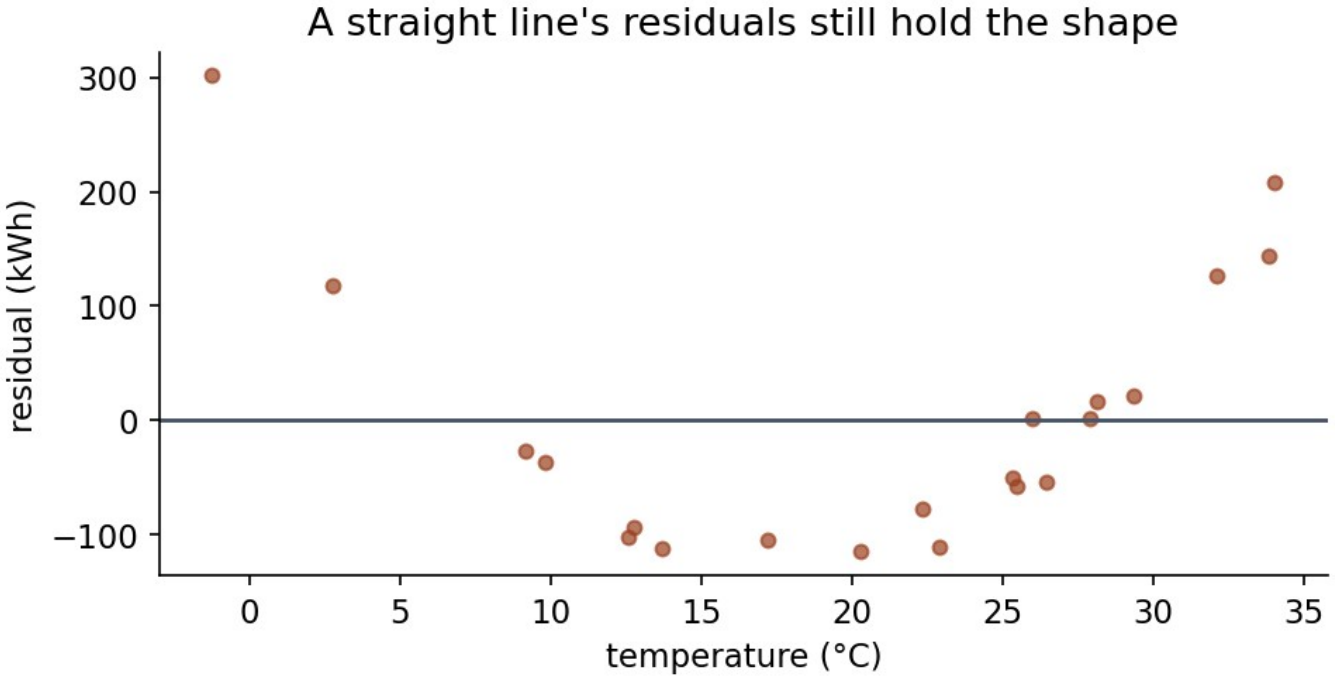
A relationship a line cannot follow



“Linear” means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

The straight line’s residuals keep the shape



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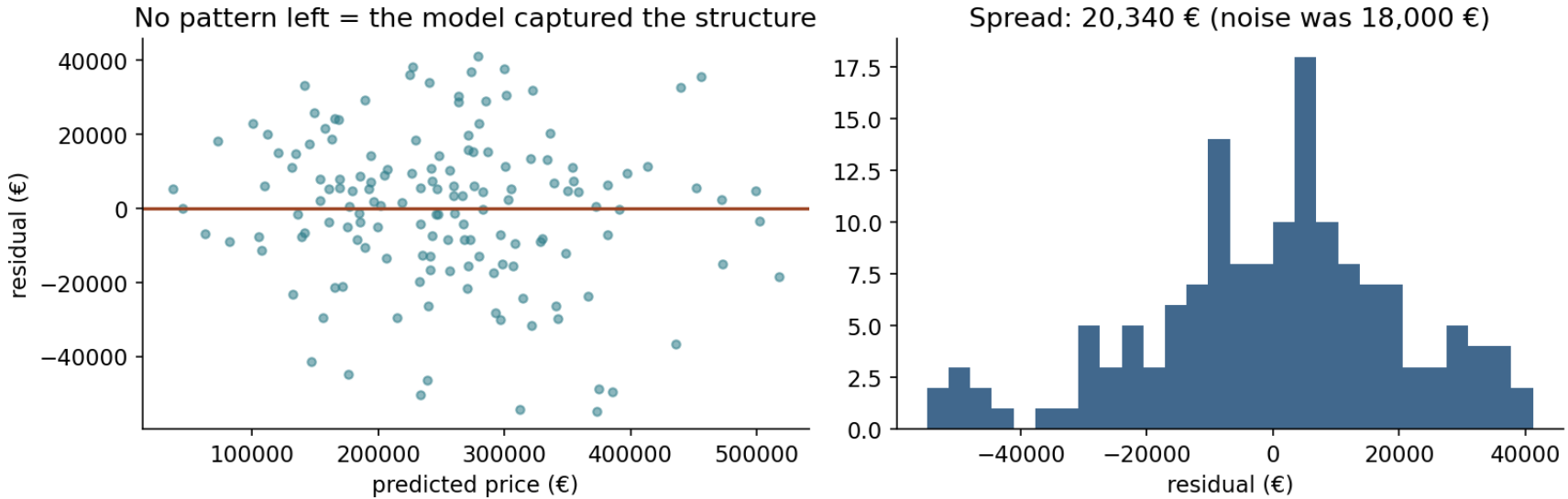
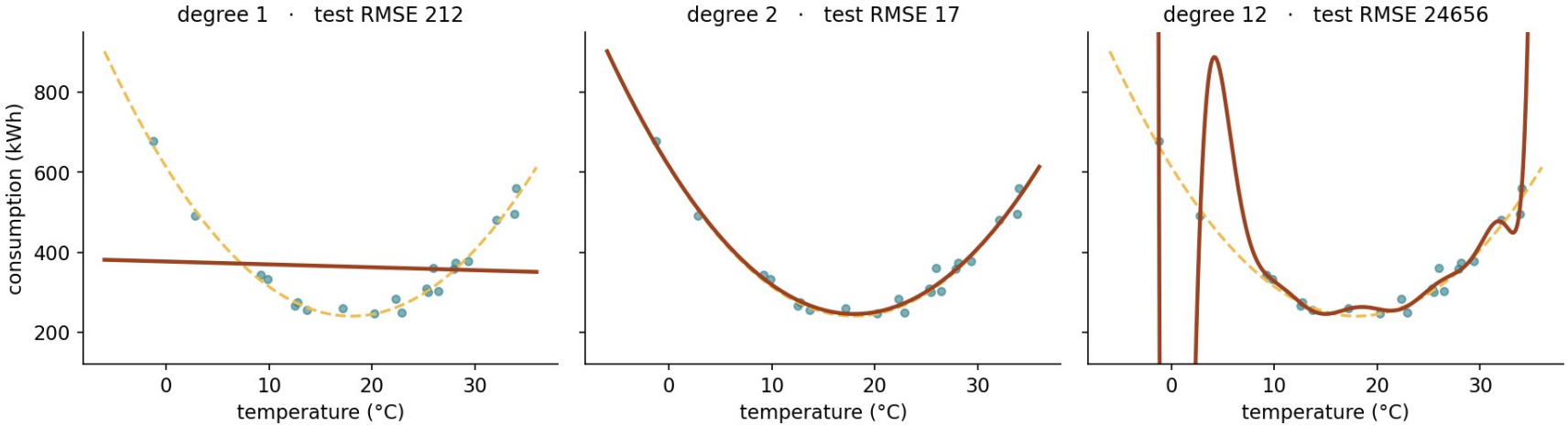
What “nothing left” looks like



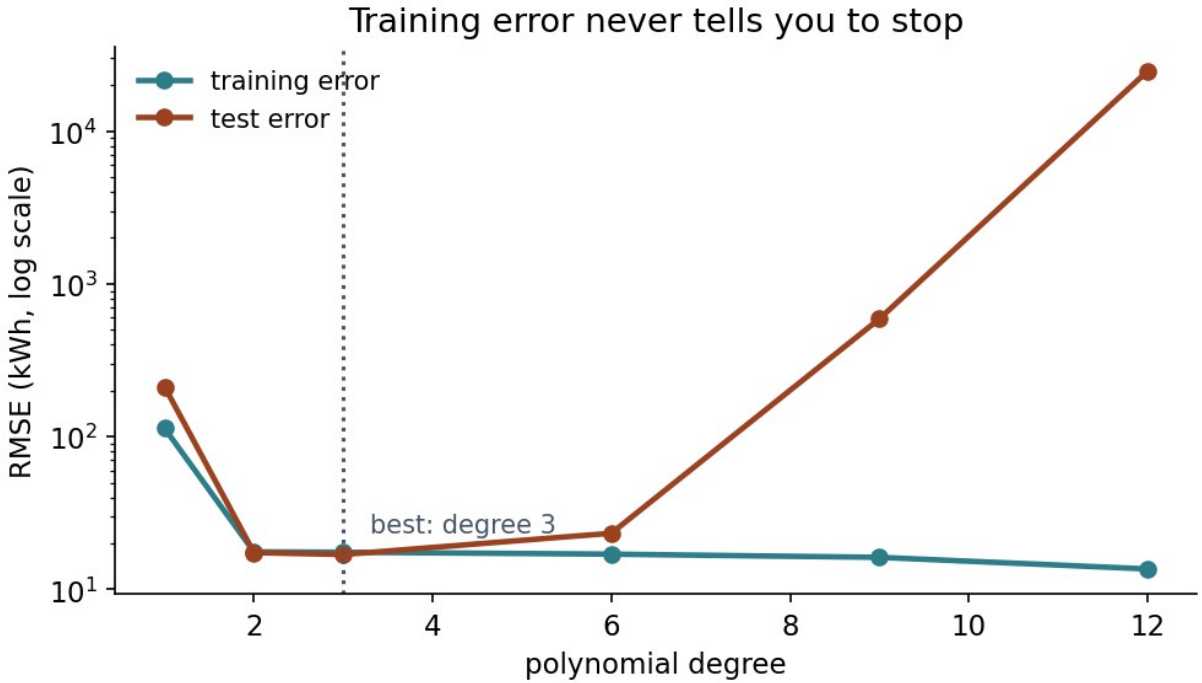
So use more flexibility?



Root mean squared error (RMSE), by degree

Degree	Train RMSE	Test RMSE
1	113.3	212.1
3	17.5	**16.9**
9	16.2	590.1
12	13.6	24,655.7

The gap is the overfitting



Notebook 2, live

Fit the curve, push the degree up, and watch the test error turn.

Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **3,097,038,010**

Charge for size

$$J_{\text{ridge}} = \text{MSE} + \lambda \sum_j w_j^2 \qquad J_{\text{lasso}} = \text{MSE} + \lambda \sum_j |w_j|$$

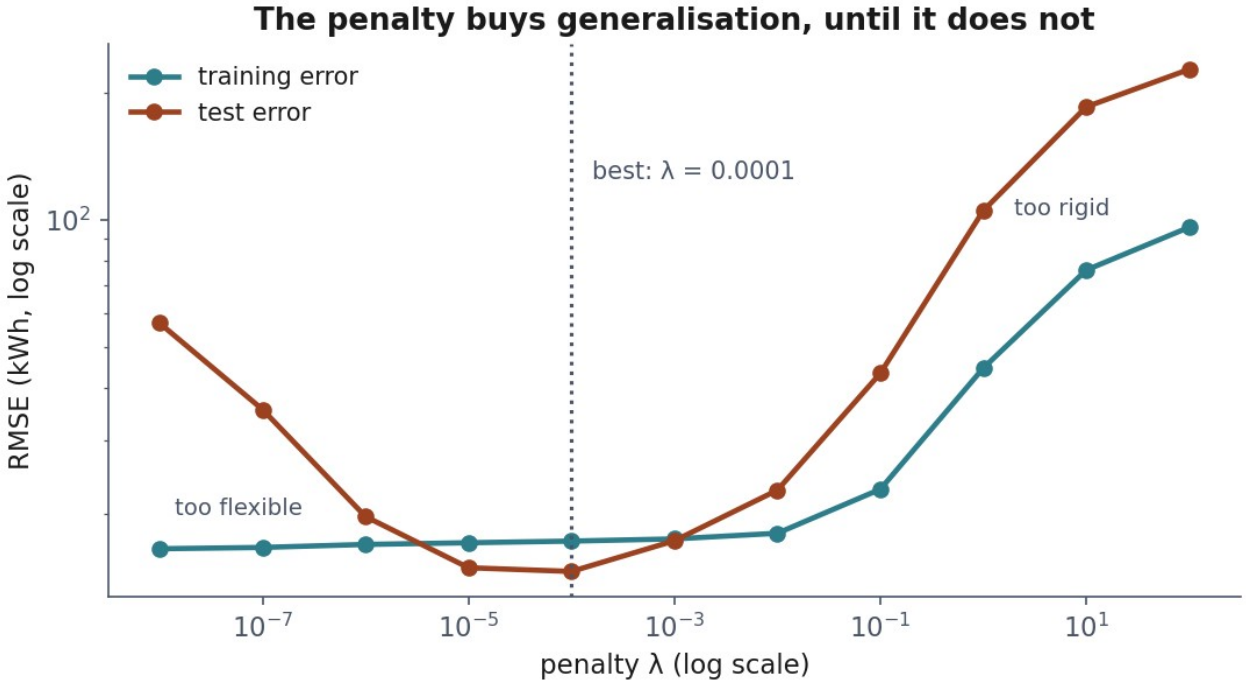
Two rules that are not optional

- The **intercept is never penalised**: it is not a claim about any feature
- Features **must be scaled first**, or the penalty is arbitrary

What a penalty buys

Model	Train	Test	max |w|
none	13.6	24,655.7	3,097,038,010
λ = 0.01	18.0	**22.8**	365
λ = 1	44.5	105.2	156
λ = 100	96.0	227.6	6

Too flexible, too rigid



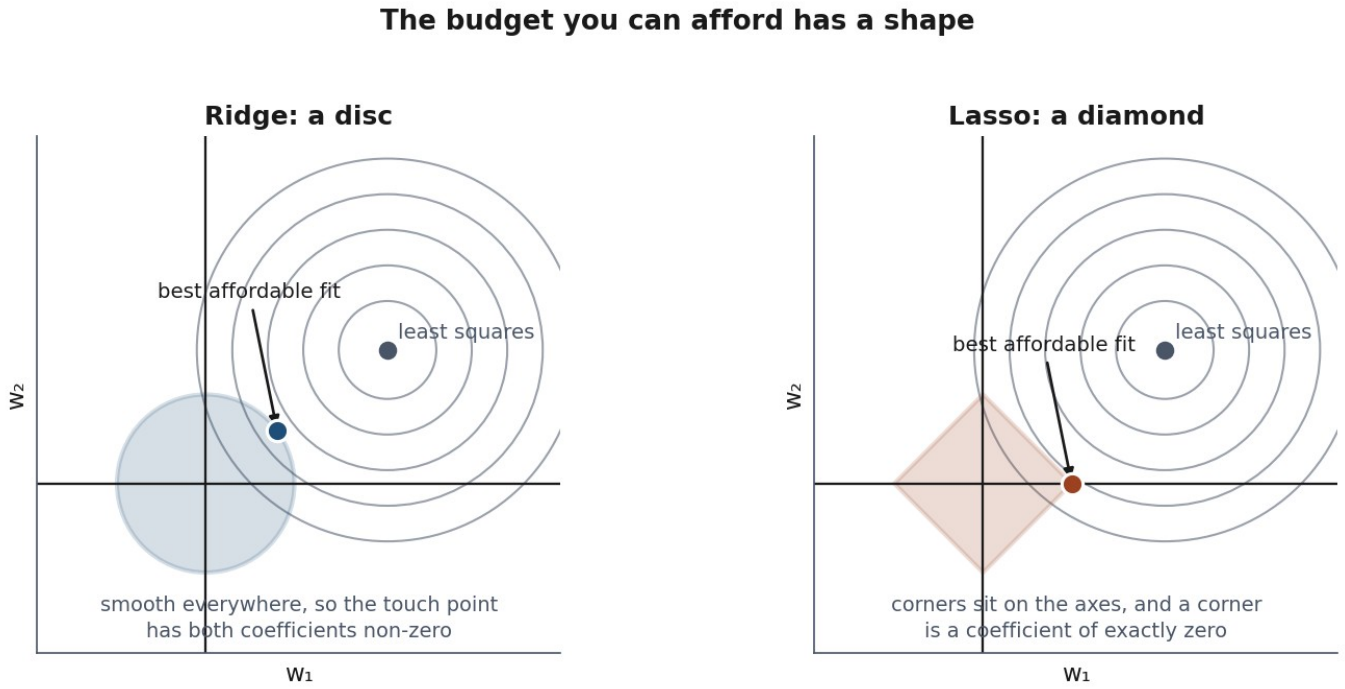
Ridge has an exact solution too

$$\theta = (X^\top X + \lambda I)^{-1} X^\top y$$

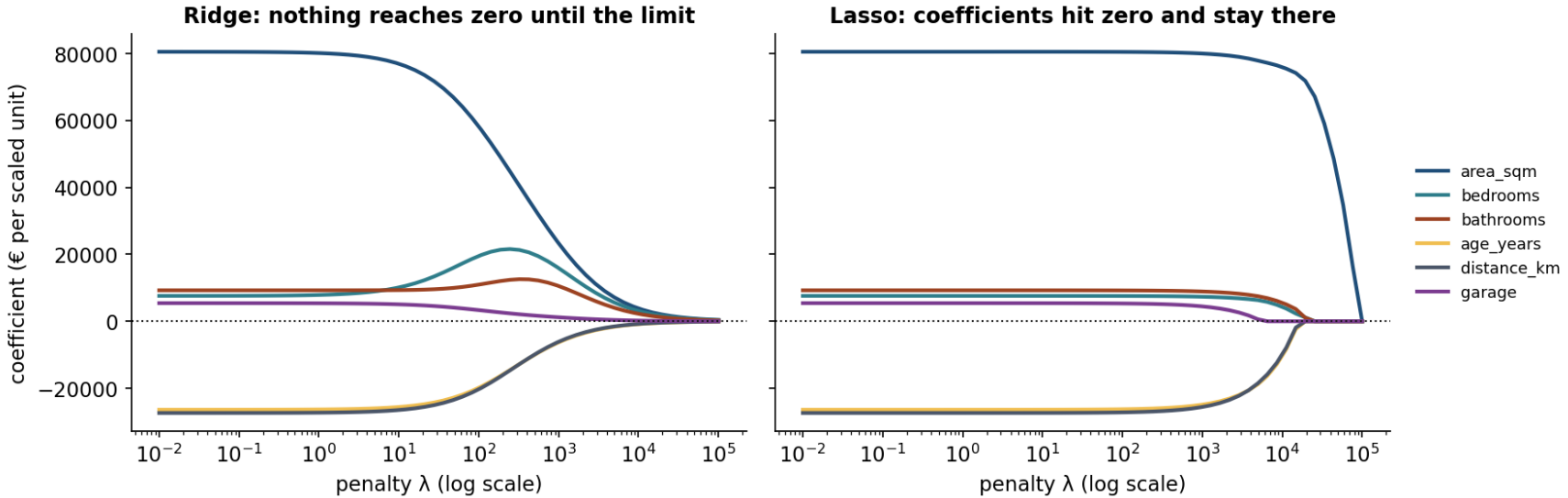
Why that one change fixes it

- Least squares fails when a direction costs nothing: the valley is flat
- The penalty makes **every** direction cost something
- Every eigenvalue shifts up by λ, so none of them can be zero

The picture behind Ridge vs Lasso



Ridge shrinks, Lasso selects



The order Lasso drops features

λ	Features kept
1,000	all six
10,000	five: `garage` dropped
20,000	three: `area`, `bedrooms`, `bathrooms`
40,000	one: `area_sqm`

The case Ridge was invented for

`area_sqm` and `area_sqft`: the same measurement, two units.

Correlation: **0.999999**

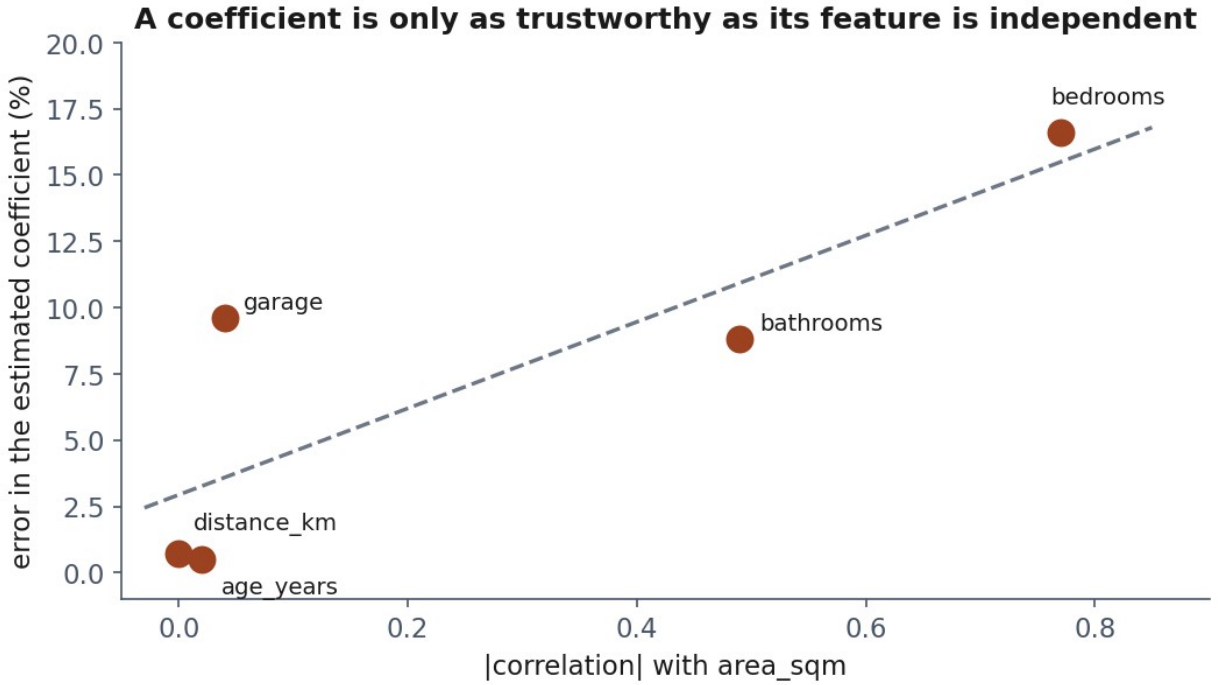
The coefficients are noise

Split	area_sqm	area_sqft
0	9,261	−640
1	14,535	−1,127
2	45,307	−3,989
3	39,061	−3,402

What a coefficient actually says

The change in the prediction for a one-unit change in that feature, **with every other feature held fixed**.

When can you trust a coefficient?



Notebook 3, live

Ridge on the disaster, the regularisation paths, the collinear case.

What we did today

- A model that claims a fixed amount per unit
- An exact solution (a projection), and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients
- **365** against billions: what a penalty of 0.01 costs, and buys

Homework: we discuss it on Friday 16 October

`Exercises/03_regression.md`

Fit, regularise, and **explain which coefficients you believe**.